



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number:01

Experiment Title: Design and Simulation of a Full Adder Circuit using Verilog HDL and Xilinx ISE

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1. Experiment Name Design and Simulation of a 1-bit Full Adder using Verilog HDL in Xilinx ISE

2. Objective

The objectives of this experiment are:

- To design a 1-bit Full Adder circuit using Verilog HDL.
- To implement the design using Xilinx ISE.
- To generate RTL schematic of the designed circuit.
- To simulate the circuit and verify Sum and Carry outputs.
- To understand FPGA-based digital circuit design flow.

3. Apparatus / Hardware & Software Requirements

Software:

- Xilinx ISE Design Suite
- ISim Simulator
- Verilog HDL

Hardware:

- Computer/Laptop system

4. Circuit Diagram / Schematic

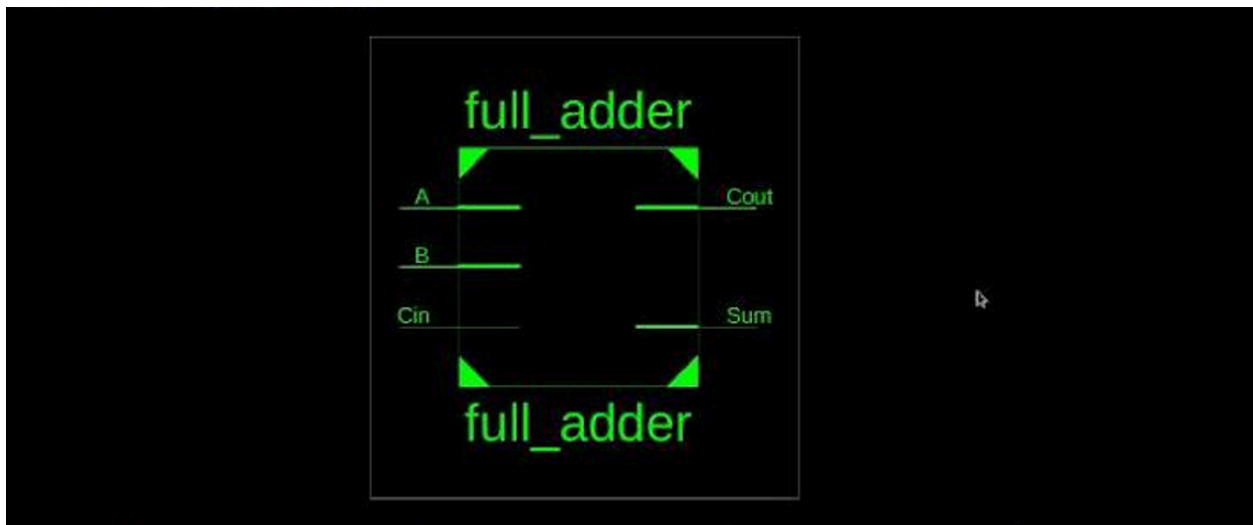


Figure 01: RTL Schematic of Full Adder

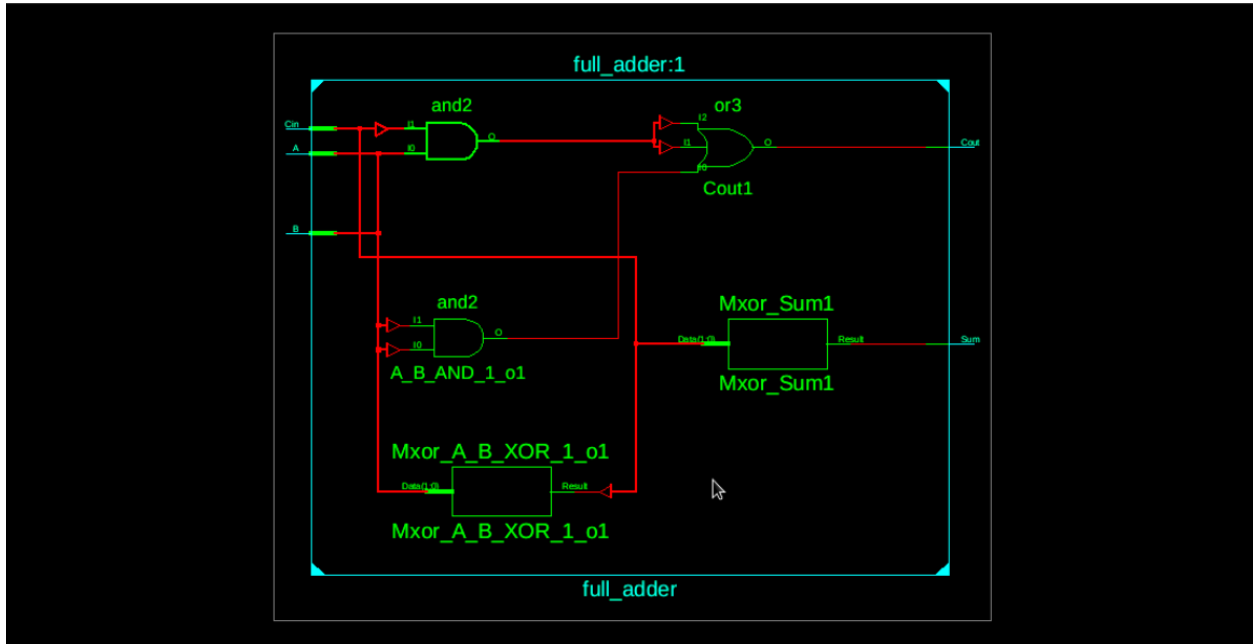


Figure 02: Gate-Level RTL Schematic of Full Adder

Figure	Schematic Type	Description
Figure 1	RTL Schematic of Full Adder	Shows the top-level Full Adder module with input ports (A, B, Cin) and output ports (Sum, Cout).
Figure 2	Gate-Level RTL Schematic of Full Adder	Shows the internal logic implementation using XOR, AND, and OR gates to generate Sum and Carry outputs.

6. Output Analysis / Observations

Simulation Result:

The Full Adder circuit was simulated using ISim simulator. All possible input combinations of A, B, and Cin were applied. The waveform verified that the generated Sum and Carry outputs matched the expected Full Adder truth table.

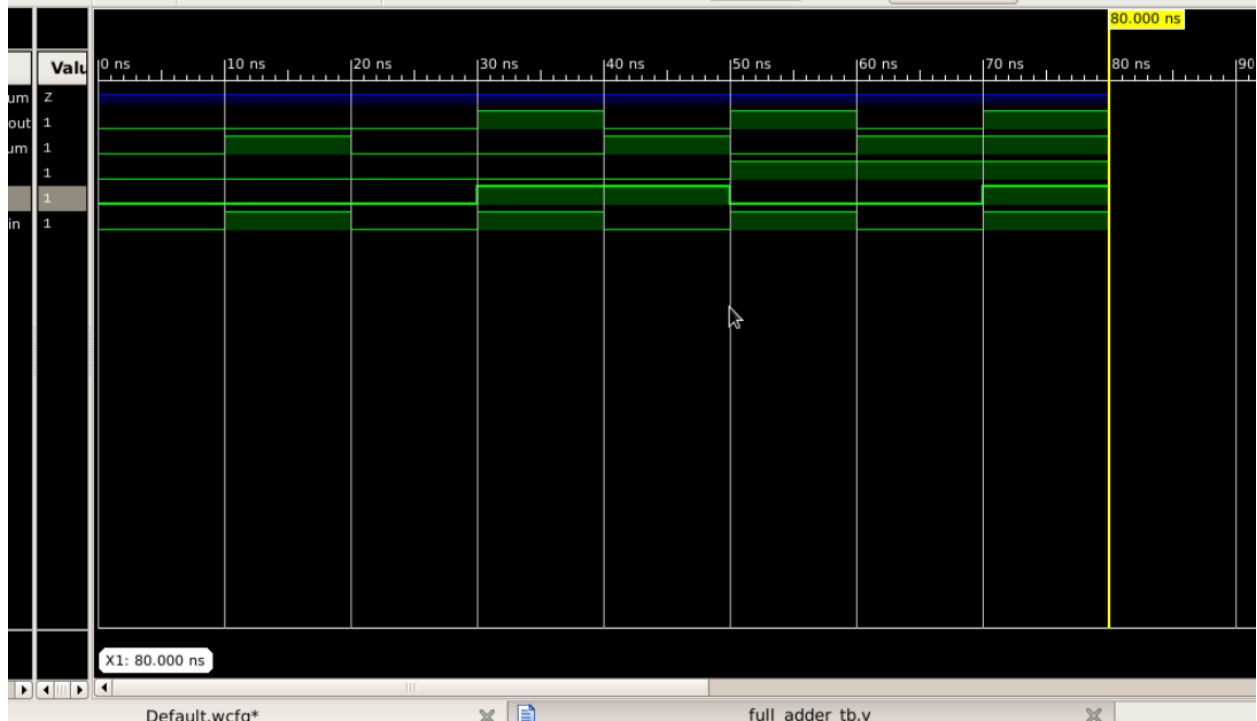


Figure 03: Behavioral Simulation Waveform of Full Adder

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 01: Truth Table of 1-bit Full Adder

The 1-bit Full Adder circuit was successfully designed and simulated using Verilog HDL in Xilinx ISE. The RTL schematic and simulation waveform verified the correct generation of **Sum** and **Carry (Cout)** outputs for all possible input combinations.

The objective of the experiment was successfully achieved by implementing the Full Adder design, analyzing its internal logic structure, and validating its functionality through simulation.

8. Github Repository Commits

GitHub Repository: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/tree/main/Full_Adder

Commit Link: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/commits/main/Full_Adder

9. Conclusion

The experiment provided practical understanding of designing and simulating a digital circuit using Verilog HDL and Xilinx ISE. The Full Adder was successfully implemented, synthesized, and verified through RTL schematic and behavioral simulation.